

Upset Losses, Compliant Citizens: Evidence from Chinese Football

Anonymous

Abstract

Do incidental emotional shocks affect political attitudes toward state authority? We link Chinese Super League football results to respondents' interview dates in the Chinese General Social Survey (2015, 2018, 2021, and 2023) and exploit within-province-year variation in exposure to unexpected match outcomes, classified against pre-match betting odds. Upset losses by a respondent's provincial team reduce support for government non-interference with public criticism by 0.294 points on a five-point scale (0.26 standard deviations) in the one-day window. The effect attenuates over the following week and is absent for close losses and upset wins. It appears larger among respondents more likely to have seen the result and is concentrated in the public-criticism item. Generalized social trust and happiness are unaffected, and neither of the two companion government-non-interference items shows a comparable response. These findings are consistent with transient negative emotions increasing acceptance of government control over public criticism, a reversal of the same-valence pattern documented in the political-spillover literature.

Keywords: incidental emotions, political attitudes, compensatory control, football, China

JEL Classification: D91, P20, Z20

1 Introduction

When a team loses, the consequences can reach well beyond the stadium: if the team was favored to win, family violence rises and juvenile sentences lengthen (Card and Dahl, 2011; Eren and Mocan, 2018); national stock markets dip the next day (Edmans et al., 2007); and even a narrow defeat can erode trust in political leaders (Kikuta and Uesugi, 2023). If a lost football match can shake citizens’ trust in those who govern them, can it also change how much authority they are willing to grant the state? Acceptance of government control over political expression is central to governance, yet little is known about what shapes it beyond the government’s own conduct. China offers a natural setting for this question: the State Council has made football’s development a national project (General Office of the State Council, 2015), and the professional league’s clubs are widely followed and closely identified with their home provinces. We ask whether an unexpected loss by a province’s football team, an emotional shock with no connection to politics, shifts citizens’ acceptance of government control over public criticism.

Behind the violence and the longer sentences lies a common trigger: not the loss itself, but the gap between the realized and the expected outcome (Kőszegi and Rabin, 2006; Card and Dahl, 2011). The same logic has been extended to consumer spending and tipping (Cai and Li, 2026; Ge, 2018). Where these shocks reach political attitudes and behavior, the documented response has so far tracked the valence of the shock (Healy et al., 2010; Kikuta and Uesugi, 2023). Whether a negative shock of this kind can instead *increase* acceptance of state authority, reversing the same-valence pattern, has, to our knowledge, never been documented.

We link Chinese Super League (CSL) match results to individual interview dates in four waves of the Chinese General Social Survey (CGSS) between 2015 and 2023. Following Card and Dahl (2011) and Ivandić et al. (2024), we classify match outcomes relative to pre-match betting odds, so that an upset loss is a defeat in a match the bookmakers expected the team to win. Identification comes from timing:

an upset is, by construction, the unanticipated part of a match outcome, so whether a respondent happens to be interviewed just after one is plausibly as good as random. We therefore compare respondents in the same province and survey year whose interviews fall shortly after an unexpected defeat with those interviewed at other times; province-by-year fixed effects absorb conditions common to a province in a given year.

Upset losses reduce support for government non-interference with public criticism by 0.294 points on a five-point scale ($p < 0.01$) within a one-day window, an effect equivalent to 0.26 standard deviations. In other words, respondents interviewed the day after their provincial team suffers an unexpected defeat are more willing to accept government intervention when someone publicly criticizes it. The reduction fades to 0.224 points ($p < 0.01$) at three days and 0.149 points ($p < 0.05$) at seven days. This sharp initial response followed by rapid decay is consistent with a transient emotional channel rather than a lasting attitude change. Close losses and upset wins produce no comparable effects, and the response appears larger among respondents more likely to have seen the result. Upset losses are rare, and the effect is identified off a small number of treated provinces, so we complement conventional inference with randomization and wild-cluster bootstrap methods (Section 6). We rest the case not on any single test but on the specificity of the response: to surprise, to one political domain, and to more-exposed respondents.

The response is also not a diffuse bad mood. A general souring of mood would spill over into whatever judgment respondents are asked to make (Schwarz and Clore, 1983). Yet the observed shift is concentrated in attitudes toward public criticism of the government. Nothing comparable appears for two companion items concerning government non-interference in childbearing or in choices of work and residence, nor does the shift extend to generalized social trust or self-reported happiness. We interpret this domain-specific response as a reaction to threatened control. An upset loss is sudden, unexpected,

and beyond control, and it offers no obvious agent to blame. That appraisal profile points to fear rather than anger, which typically arises when a negative outcome is attributed to a blameworthy agent (Smith and Ellsworth, 1985; Lerner and Keltner, 2001). When control is threatened and no clear target of blame is available, individuals may restore a sense of order by bolstering the external systems that provide it, the government among them, a response known as compensatory control (Kay et al., 2008, 2009). Among the domains we examine, public criticism of the government most directly challenges the authority of this order-providing institution, which may explain why it is the only domain in which attitudes respond to an upset loss.

We contribute to two literatures. First, to the literature on incidental emotional shocks and their political spillovers, we add the missing direction: a negative shock that makes citizens *more*, not less, likely to accept state authority. In the political cases documented so far, the direction of the response has generally followed the valence of the sporting shock: pre-election wins raise incumbent vote shares (Healy et al., 2010), and close losses lower trust in national leaders (Kikuta and Uesugi, 2023). To our knowledge, we provide the first field evidence of the opposite pattern: an incidental negative shock, in which the state plays no part, increases acceptance of government control over public criticism. This finding does more than add another instance of political spillover from sports. It indicates that the effect of a negative emotion on political attitudes depends less on the valence of the shock itself than on the attitude it meets, consistent with evidence that induced negative emotions elsewhere shift policy views without raising acceptance of authority (Algan et al., 2026). Its direction is also the one compensatory-control theory predicts for an outcome that concerns social order.

Second, to the study of political attitudes in China, we add a determinant that lies outside the government's own conduct. Existing work examines how citizens' trust in and support for the state respond to government performance and political information (Fang et al., 2025), how the state manages political

expression (King et al., 2013), and how broadly favorable public opinion toward the central government moves with material conditions (Cunningham et al., 2020). We show that attitudes toward government control over public criticism also respond to incidental emotional cues orthogonal to government performance. The demand side of political control, the degree of authority citizens are willing to grant the state, may therefore be more malleable in the short run than accounts centered on instrumental evaluation or stable cultural values would suggest. A practical corollary is that short-run variation in measured political attitudes may reflect not only what the government has done, but also when citizens happen to be surveyed.

The remainder of the paper proceeds as follows. Section 2 discusses background and motivation. Section 3 describes the research design. Section 4 presents the main results and the mechanism evidence. Section 5 examines identification through event-study and balance tests. Section 6 reports robustness and inference. Section 7 concludes.

2 Background and Motivation

Three literatures jointly motivate our test and make it possible: sporting upsets provide credible incidental emotional shocks; the documented political responses to such shocks generally mirror their valence; and the psychology of threatened control identifies the conditions under which that pattern should reverse. Our design belongs to a lineage that treats sporting outcomes as incidental emotional shocks, isolated not by the result itself but by an expectation benchmark (Kőszegi and Rabin, 2006): an upset is defined against point spreads (Card and Dahl, 2011; Eren and Mocan, 2018; Cai and Li, 2026), betting odds (Ivandić et al., 2024; Ge, 2018), or rankings (Rees and Schnepel, 2009). The emotional cue is therefore orthogonal to whatever the benchmark already prices, including team quality, form, and the stakes of the match. In the violence and sentencing domains the response follows the valence of

the surprise, and negative surprises dominate (Card and Dahl, 2011; Cardazzi et al., 2024; Eren and Mocan, 2018); asset prices fall after losses irrespective of pre-match expectations (Edmans et al., 2007); the exceptions are themselves informative, with upset losses raising retail-therapy spending (Cai and Li, 2026) and tips responding to surprise wins alone (Ge, 2018). Already within these behavioral domains, the direction of the response is not fixed by the valence of the shock alone.

The lineage rests on an individual-level microfoundation, and it has a known scope condition. Kessler et al. (2022) supply the former, instrumenting within-person happiness with live game events in a sports-bar experiment and moving incentivized decisions over charitable giving and trust.¹ The scope condition comes from German league matches: match-day violence tracks rivalry and group identity while odds-based unexpected outcomes contribute nothing (Andres et al., 2023). Table 1 organizes this evidence by shock, identification, and signed response; across these settings, whether surprise is the operative cue is an empirical question rather than an assumption.

Because upsets are rare events in observational data, the literature polices itself against false positives. Fowler and Montagnes (2015) reexamine the electoral result of Healy et al. (2010) and argue it is spurious: the estimate is no stronger where college football commands more attention, is undiminished where the mechanism predicts it should weaken (with no incumbent on the ballot, or outside the team’s home county), and has no analogue in the far more popular NFL. We read the episode as setting the evidentiary bar for this design family rather than discrediting it. It asks that shocks be classified against market-priced expectations, that placebo outcomes bound what the shock is allowed to move, and that inference respect the small number of treated units; Sections 3, 4, and 6 implement each of these standards. Replication across settings matters for the same reason, and the same-valence political spillover does recur: national-team victories shift self-identification from the ethnic group toward the nation in

¹Coviello et al. (2024) use the same class of instrument, local team results, for worker mood and find that *better* mood lowers output among fixed-wage call-center workers, a reminder that what a mood shift does depends on the margin it meets.

Table 1: Incidental emotional shocks and the direction of their spillovers.

Study	Shock (identification)	Response
<i>Panel A: Behavioral and economic outcomes</i>		
Card and Dahl (2011)	Upset NFL losses, US (point spread)	Family violence rises
Edmans et al. (2007)	International soccer losses (effect unrelated to pre-match expectations)	Stock returns fall
Eren and Mocan (2018)	Upset college football losses, US (point spread)	Juvenile sentences lengthen
Cai and Li (2026)	Unexpected NFL outcomes, US (point spread)	Spending rises after losses, falls after wins
<i>Panel B: Political outcomes</i>		
Healy et al. (2010)	Pre-election college football wins, US	Incumbent vote share rises
Depetris-Chauvin et al. (2020)	National-team victories, sub-Saharan Africa (interview timing; close qualification)	Ethnic identification falls; civil conflict falls
Kikuta and Uesugi (2023)	Close losses, European clubs fielding African players (interview timing; odds-conditioned)	Trust in leaders falls; peaceful protest rises
Guo et al. (2025)	Female Olympic gold medals, China (DiD across CGSS waves)	Desire for daughters rises
Chen and MacDonald (2020)	Broadcast of athletic achievements, China (randomized)	Local-government satisfaction and compliance rise
<i>Panel C: Non-sports incidental affect</i>		
Algan et al. (2026)	Induced negative emotions, US (survey experiments)	Protectionism and redistribution support rises; populist attitudes unchanged
Fochmann et al. (2025)	Induced positive emotions, Germany (laboratory)	Tax compliance falls
Meier et al. (2019)	Voting-weekend rainfall, Switzerland	Votes for political change fall

Notes: Each row is a representative study; where a parenthetical appears, it states the device isolating the unexpected or exogenous component of the shock. Related studies that replicate or qualify these designs are discussed in the text.

sub-Saharan Africa ([Depetris-Chauvin et al., 2020](#)), and close losses by European clubs fielding African players depress trust in national leaders across African countries ([Kikuta and Uesugi, 2023](#)). Where the design family reaches political outcomes and clears this bar, then, the response has so far always shared the sign of the shock.

Evidence from China bears on each element of our design, but no study runs the test itself. Emotional cues from sports do shift attitudes in China: [Chen and MacDonald \(2020\)](#) randomize exposure to a broadcast of Chinese athletic achievements and find higher satisfaction with and compliance toward local government. Using the CGSS as we do, [Guo et al. \(2025\)](#) show that the Rio 2016 gold medals of female champions raised the desire for daughters in their home provinces across survey waves. Both studies examine favorable events. For a negative cue, the closest natural experiment is the Wenchuan earthquake: political trust rises among respondents interviewed just after the quake, and the rise dissipates within weeks ([You et al., 2020](#)). That event, however, bundles the shock with the state’s own highly visible relief mobilization and media rally, so it captures, at least in part, citizens’ evaluations of government in action. Isolating incidental emotion requires a shock in which the state plays no part, which club football provides. On the outcome itself, attitudes toward the state in China are often modeled as instrumental evaluations of government performance ([Fang et al., 2025](#); [Cunningham et al., 2020](#)), while direct questions on sensitive items may overstate support because of self-censorship ([Robinson and Tannenber, 2019](#)), a measurement concern that a shock-based design can difference out (Section 3). What remains unanswered is whether an emotional cue orthogonal to government action moves expression-related attitudes at all.

Two gaps follow from Table 1. First, every political spillover from an incidental shock, Panel B in full, runs in the same direction as the shock; the departures visible elsewhere in the table lie outside the political panel: retail-therapy spending after upset losses in Panel A and the non-sports patterns of

Panel C. Second, none of the outcomes concerns acceptance of state limits on political expression. That the valence mapping is outcome-dependent is precisely what the non-sports evidence shows: induced negative emotions raise support for protectionist and redistributive policies in the United States without hardening attitudes toward authority ([Algan et al., 2026](#)), incidental positive emotions lower tax compliance ([Fochmann et al., 2025](#)), and rain-induced negative affect shifts referendum votes toward the status quo ([Meier et al., 2019](#)). In these studies the direction of the spillover tracks what the emotion disposes people to do (seek protection, hold to the familiar, or let compliance slip), rather than the valence of its trigger. No row of Table 1 pairs a negative incidental shock with rising acceptance of state authority: to our knowledge, that combination, acceptance of state limits on public criticism after an incidental negative shock, remains undocumented.

Appraisal theory explains why a reversal should emerge here: incidental emotions carry their appraisal tendencies, not merely their valence, into unrelated judgments ([Lerner and Keltner, 2001](#)). An upset loss is sudden, expectation-violating, and beyond control, and it provides no culpable agent. This appraisal profile is associated with anxiety and threatened control ([Smith and Ellsworth, 1985](#)), rather than with anger, which is organized around perceived unfairness and an identifiable target of blame, the configuration that theories of political unrest link to anti-government action ([Passarelli and Tabellini, 2017](#)). Compensatory-control theory predicts how people respond when their sense of control is threatened but no clear target of blame is available: they seek to restore a sense of order by bolstering external systems that can provide it, with greater support for government among the best-documented responses ([Kay et al., 2008, 2009](#); [Landau et al., 2015](#)). Consistent with this mechanism, uncertainty and threat raise the appeal of strong, order-providing authority ([Kakkar and Sivanathan, 2017](#)), with perceived threat activating authoritarian predispositions ([Feldman and Stenner, 1997](#)). For this compensatory turn to favor the state, the state must be regarded as a credible provider of order. That premise is plausibly

met in China, where approval of the central government is persistently high and moves with performance (Cunningham et al., 2020), and local governments respond to citizen demands (Chen et al., 2016), complementing institutional accounts of regime resilience (Nathan, 2003). The resulting prediction is precise: heightened concern for order should be directed specifically at the behavior perceived to threaten order, public criticism of the government, not at government control in general. The placebo outcomes in Section 4 are built to detect exactly this specificity. No existing study has run this test on expression-related attitudes with an incidental shock in China; addressing this gap is the central contribution of our paper.

3 Research Design

3.1 Institutional Setting: The Chinese Super League

The Chinese Super League (CSL) is the top tier of professional football in China. Founded in 2004, the league comprises 16 clubs, each based in a provincial capital or major city and closely identified with its home province. In March 2015, the General Office of the State Council issued the Overall Plan for Chinese Football Reform and Development, a national reform blueprint for the game (General Office of the State Council, 2015; Cockayne et al., 2022). Much of our sample falls in the league’s *jinyuan* (“money football”) boom years, when clubs backed by property developers and state-linked owners recruited established international stars: they broke the Chinese transfer record five times in 2016 alone (Church, 2016), with Chelsea’s Oscar arriving for 60 million pounds (Twomey, 2016). The boom showed in the stands and on screens: by the 2015 season the league averaged above 21,800 spectators a match, within a few hundred of Serie A and Ligue 1 (Hennelly, 2016), and its five-year broadcast rights for 2016–2020 sold for a record 8 billion yuan, about 1.25 billion USD (Xinhua, 2015). The combination of strong provincial identification with local clubs, star players, and broad media reach makes CSL outcomes a plausible source of short-lived emotional shocks for the general provincial population.

3.2 Data

Football data Following Card and Dahl (2011) and Ivandić et al. (2024), we use betting markets to measure pre-match expectations and classify match outcomes into emotional-cue categories. Pre-match odds define the benchmark against which fans evaluate realized outcomes (Kőszegi and Rabin, 2006), so a defeat when the team was favored generates a larger negative response than one that was expected (Card and Dahl, 2011). We collect match-level data for all CSL games from 2014 to 2023 from www.football-data.co.uk. The data provide the match date, home and away teams, full-time result (home win, draw, or away win), and pre-match decimal odds for each of the three outcomes. We additionally scrape match-level home attendance and club net transfer spending for all CSL clubs from Transfermarkt, which enter the robustness analysis as controls (Figure 6).

We convert odds into implied probabilities and remove the bookmaker margin by normalizing inverse odds so that $p_H + p_D + p_A = 1$. Using the team’s implied win probability p_{win} , we classify each match from each team’s perspective as a *predicted win* ($p_{\text{win}} \geq 0.55$), *predicted loss* ($p_{\text{win}} < 0.45$), or *predicted close* (otherwise). These thresholds follow Ivandić et al. (2024), who apply the same cutoffs to European football leagues.²

Combining these predicted-outcome bins with realized results yields six mutually exclusive categories: *upset loss* (expected win, actual loss), *close loss* (close match, actual loss), *predicted loss* (expected loss, actual loss), and the corresponding win-side categories *upset win*, *close win*, and *predicted win*. Draws form a separate category and do not enter the win/loss classifications. An upset loss thus captures the negative surprise of a defeat when the team was favored, while a predicted loss carries little new information and should produce a weaker response (Kőszegi and Rabin, 2006; Card and Dahl,

²Card and Dahl (2011) classify NFL games using a point-spread threshold of 4 points. Because football (soccer) betting markets quote three-way odds rather than point spreads, we adopt the probability-based classification of Ivandić et al. (2024), which maps onto an analogous distinction between games where the team is a clear favorite, games that are roughly evenly matched, and games where the team is a clear underdog.

2011).

Figure 1 plots the realized goal difference (home goals minus away goals) against the implied home-win probability for all CSL matches in our sample. The positive relationship validates betting odds as informative measures of pre-match expectations.

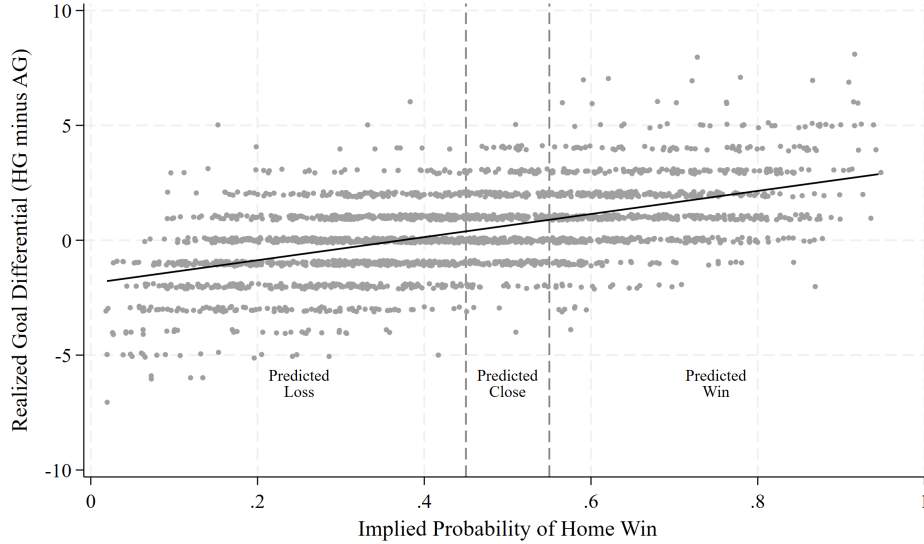


Figure 1: Betting market efficiency.

Notes: Each point is a Chinese Super League match, 2014 to 2023. The horizontal axis is the implied home-win probability from normalized betting odds (football-data, Pinnacle closing); the vertical axis is the realized goal difference (home minus away). The solid line is a linear fit; the vertical dashed lines mark the $p = 0.45$ and $p = 0.55$ thresholds separating predicted losses, close matches, and predicted wins.

Household survey data We draw individual-level data from the Chinese General Social Survey (CGSS), a nationally representative repeated cross-section administered by the National Survey Research Center at Renmin University of China. The CGSS uses a multi-stage stratified probability sampling design covering all provinces in mainland China, and records the exact date of each face-to-face interview, which is essential for linking respondents to recent match outcomes in their province. We pool the 2015, 2018, 2021, and 2023 waves, which share a common set of attitudinal and demographic items.³

³We exclude the 2017 wave, which does not record interview dates and therefore cannot be linked to match outcomes. The 2021 wave measures social trust on a four-point rather than five-point scale; accordingly, social trust regressions use only the

Respondents rate their agreement with the statement that “the government should not interfere when someone publicly criticizes it,” on a scale from 1 (strongly disagree) to 5 (strongly agree), which we label *Non-interference: public criticism*. A respondent who fully agrees scores a 5, the strongest support for non-interference, so a negative treatment coefficient reflects greater acceptance of government intervention against public criticism. We interpret the item as measuring tolerance of state restrictions on public criticism, rather than as a broad evaluation of the government or of freedom of expression in the abstract. In the terminology of the political-tolerance literature, it is a tolerance-of-dissent item in which the acting party is the state rather than the respondent: the question is whether the government should refrain from acting against a critic, not whether the respondent would extend expressive rights to speech they dislike. Because the item does not specify a particular level of government, we interpret responses as referring to government authority in general. This construct connects to research on how the Chinese state manages political expression (King et al., 2013) and to the compensatory-control account we adopt, in which a threatened sense of personal control lowers willingness to tolerate criticism of the government (Kay et al., 2009). Direct questions on politically sensitive items can also overstate expressed support through self-censorship (Robinson and Tannenbergs, 2019). Our design mitigates this concern by exploiting the as-if-random timing of an emotional shock: any stable tendency to misreport should be differenced out across exposed and unexposed respondents and therefore cannot generate a response confined to one item and one short window. To gauge the specificity of this response, we also examine four placebo outcomes that should be unmoved if the effect is a domain-specific reaction to political criticism: two other government-non-interference items, and two general-mood measures.⁴

Individual-level controls include gender, Han ethnicity, years of education, marital status, Chinese 2015, 2018, and 2023 waves.

⁴The two additional government-non-interference items ask for agreement, on the same 1–5 scale (higher indicating greater support for non-interference), that “how many children to have is a personal matter, and the government should not interfere” and that “where to work and live is a personal freedom, and the government should not interfere.” The two mood measures are generalized social trust and life happiness, each on a 1–5 scale on which higher values indicate, respectively, greater trust and greater happiness. All four are summarized in Table 2 and used in Figure 3.

Communist Party membership, agricultural *hukou* status, self-rated health, and the log of personal income. These variables capture the key demographic and socioeconomic characteristics that prior work has shown to predict political attitudes and subjective wellbeing in China (Fang et al., 2025; Zhang et al., 2025; Asadullah et al., 2018; Zhang and Churchill, 2020). We additionally control for television and internet use frequency (each on a 1–5 scale) to account for differential media exposure that might correlate with both awareness of match outcomes and attitudinal responses.

Province-level variables We do not include province-level covariates such as GDP per capita, population, or industrial structure in the main regressions, because the province-by-year fixed effects absorb all province-year variation, including any contemporaneous economic conditions that could correlate with both match outcomes and political attitudes.

3.3 Descriptive Statistics

Table 2 reports descriptive statistics for the estimation sample. On average, respondents score 2.55 on the five-point scale of support for government non-interference with public criticism, indicating that the typical response lies between “somewhat disagree” and “neither agree nor disagree.” The football shock variables in Panel B show that upset losses are rare events in a one-day window (mean = 0.002), which motivates our use of short pre-interview windows spanning up to one week in the main analysis. Individual characteristics in Panel C are broadly consistent with the composition of the adult survey sample: 50% female, 92% Han, and roughly nine years of schooling on average.

Figure 2 illustrates the geographic distribution of our key variables. Panel (a) shows the average annual number of upset losses by province, concentrated in provinces with active CSL teams, primarily in eastern and central China. Panel (b) shows the province-level mean of support for government non-interference with public criticism. The maps highlight two features of the empirical setting. First, there

Table 2: Summary statistics.

Variable	N	Mean	Std. Dev.	Min	Max
<i>Panel A: Outcome variables</i>					
Non-interference: public criticism (1–5)	35,114	2.551	1.126	1	5
Non-interference: childbearing (1–5)	34,856	2.856	1.223	1	5
Non-interference: work/residence (1–5)	34,855	3.736	1.104	1	5
Social trust (1–5)	28,059	3.512	0.986	1	5
Life happiness (1–5)	32,752	3.899	0.818	1	5
<i>Panel B: Football shock variables (1-day window)</i>					
Upset loss	35,114	0.002	0.040	0	1
Close loss	35,114	0.001	0.029	0	1
Upset win	35,114	0.012	0.109	0	1
Predicted win	35,114	0.017	0.132	0	2
Close match	35,114	0.009	0.092	0	1
Predicted loss	35,114	0.060	0.244	0	2
<i>Panel C: Individual controls</i>					
Female (=1)	35,114	0.497	0.500	0	1
Han ethnicity (=1)	35,114	0.923	0.267	0	1
Education (years)	35,114	9.445	4.581	0	19
Married (=1)	35,114	0.749	0.434	0	1
CCP member (=1)	35,114	0.116	0.320	0	1
Agricultural hukou (=1)	35,114	0.563	0.496	0	1
Self-rated health (1–5)	35,114	3.599	1.074	1	5
Log personal income	35,114	8.410	3.943	0	16
<i>Panel D: Media controls</i>					
TV use frequency (1–5)	35,114	3.482	1.183	1	5
Internet use frequency (1–5)	35,114	3.194	1.681	1	5

Notes: Estimation sample for the main regressions: CGSS respondents from the 2015, 2018, 2021, and 2023 waves with non-missing individual controls, media controls, and sampling weights ($N = 35,114$). Social trust is available only for 2015, 2018, and 2023 (the 2021 wave uses a non-comparable four-point scale). Football shock variables are counts for the single day before the interview ($d - 1$). All statistics use CGSS sampling weights.

is substantial cross-province variation in the attitudinal outcome, which is absorbed by the province-by-year fixed effects. Second, the provinces most exposed to football shocks (e.g., Shandong, Guangdong, and Hebei) span a range of attitudinal levels, mitigating concerns that the results are driven by a single type of province.

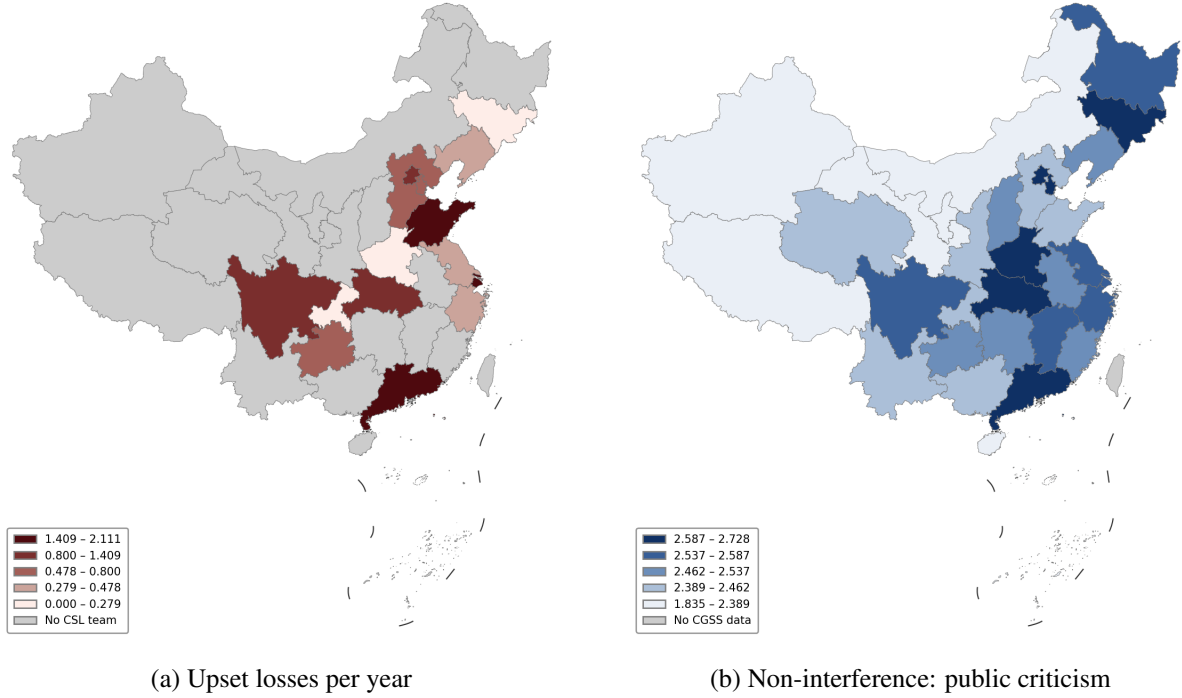


Figure 2: Geographic variation in upset losses and support for government non-interference.

Notes: Province-level maps of China. Panel (a) shows the average annual number of upset losses (unexpected defeats based on pre-match betting odds) by province, 2015 to 2023. Panel (b) displays the province-level weighted mean of the *Non-interference: public criticism* item (1 to 5 scale; higher values indicate greater support for non-interference) from the CGSS. Gray provinces have no CSL team in Panel (a) and no CGSS data in Panel (b). The base map is the standard map of China (approval number GS(2024)0650) from the Ministry of Natural Resources' National Platform for Common Geospatial Information Services (Tianditu); boundaries unmodified.

3.4 Empirical Strategy

We exploit quasi-random variation in the timing and outcomes of CSL matches relative to CGSS interview dates within the same province and year. Provinces with CSL teams differ systematically from those without, but the exact date on which a team plays, and whether it wins or loses relative to expec-

tations, are plausibly orthogonal to the date on which a given respondent is interviewed, conditional on province-by-year fixed effects. This design draws on the logic of the Unexpected Event during Survey Design (UESD) framework (Muñoz et al., 2020), which exploits the as-if-random timing of an unexpected event relative to survey interviews; we implement it as the province-by-year design below. We estimate:

$$Y_{ipt} = \beta_1 \text{UpsetLoss}_{p,W} + \beta_2 \text{CloseLoss}_{p,W} + \beta_3 \text{UpsetWin}_{p,W} + \gamma' \mathbf{C}_{p,W} + \delta' \mathbf{X}_{ipt} + \alpha_{pt} + \varepsilon_{ipt}, \quad (1)$$

where Y_{ipt} is support for government non-interference with public criticism for individual i in province p interviewed in year t . The variables $\text{UpsetLoss}_{p,W}$, $\text{CloseLoss}_{p,W}$, and $\text{UpsetWin}_{p,W}$ count the number of upset losses, close losses, and upset wins experienced by the province's CSL team(s) in the W -day window preceding the interview, specifically over the interval $[d - W, d - 1]$, where d is the interview date (we suppress d in the shock-count notation). We exclude the interview day itself to ensure that only pre-interview shocks enter the specification. The omitted category comprises respondents whose provincial team experienced none of the three shock outcomes during the window (its matches, if any, ended only in predicted wins, close wins, predicted losses, or draws). Following Card and Dahl (2011), we include predicted-outcome controls $\mathbf{C}_{p,W}$, consisting of counts of predicted wins, all close matches, and predicted losses within the same window. These controls ensure that the shock coefficients β_1 to β_3 capture the effect of deviations from expected outcomes rather than the match schedule itself. With total close matches held constant, β_2 captures the effect of a close loss relative to a close win.

The vector $\mathbf{X}_{ipt}$ includes individual-level controls: gender, Han ethnicity, years of education, marital status, CCP membership, agricultural *hukou* status, self-rated health, log personal income, and television and internet use frequency. The term α_{pt} denotes province-by-year fixed effects, which absorb all time-invariant province characteristics as well as province-specific annual factors that may correlate with

team performance and respondents' attitudes. Identification therefore relies entirely on within-province-year variation: we compare respondents interviewed in the same province and survey year but on dates that differ in their exposure to recent unexpected match outcomes. Standard errors are clustered at the province level. All regressions are weighted using CGSS sampling weights.

The coefficient of primary interest is β_1 . If an upset loss elicits negative arousal that increases demand for order and external control, we expect $\beta_1 < 0$: an upset loss should reduce support for the proposition that the government should not interfere with public criticism, i.e., increase acceptance of government intervention. The coefficient β_2 on close losses serves as a benchmark: these are losses from roughly evenly matched games that carry little surprise and should therefore produce a weaker emotional response. The coefficient β_3 on upset wins tests whether positive surprises generate comparable short-run spillovers.

We estimate equation (1) across three window widths, $W \in \{1, 3, 7\}$ days. Shorter windows ($W = 1, 3$) isolate acute responses to recent match outcomes, while the 7-day window allows us to examine whether these effects persist within the following week. The comparison across windows is informative about the nature of the effect: a sharp response concentrated in the shortest windows is consistent with a transient emotional cue rather than a durable change in attitudes (Card and Dahl, 2011).

Identifying assumption. The validity of our approach requires that, conditional on province-by-year fixed effects, the timing and outcomes of CSL matches are uncorrelated with unobserved determinants of individual attitudes. Two features of our setting support this assumption. First, the CSL schedule is set by the Chinese Football Association before the season begins and is not influenced by provincial socioeconomic conditions or survey logistics. Second, CGSS interview dates are determined by fieldwork protocols, including interviewer availability, respondent scheduling, and sampling sequences, which are plausibly independent of match outcomes. We probe this assumption in several ways in Section 5: an

event-study specification with placebo leads, which replaces the pre-interview window $[d - W, d - 1]$ with the post-interview window $[d + 1, d + W]$ and in which significant effects of future match outcomes on current attitudes would indicate confounding pre-trends or spurious seasonality, and a balance comparison of respondents interviewed just before and just after an upset.

4 Results

4.1 Main Results

Table 3 reports estimates of equation (1) across three pre-interview windows. In the one-day window (column 1), an upset loss reduces support for non-interference by 0.294 points on the five-point scale ($p < 0.01$), equivalent to 0.26 standard deviations. The effect attenuates but remains significant at three days ($-0.224, p < 0.01$) and seven days ($-0.149, p < 0.05$). This decay pattern indicates a short-run emotional channel: an incidental-emotion spillover into unrelated judgments (Lerner et al., 2015) that dissipates quickly (Card and Dahl, 2011). Because the estimate is an intent-to-treat effect at the province level, averaging over many residents who do not follow the local team, the effect on attentive fans is likely substantially larger.

Two features temper how much weight the analytic p -values can carry. Because upset losses are rare, the number of treated clusters is small: two provinces are exposed within the one-day window and four within the three- and seven-day windows. With so few treated clusters, cluster-robust tests are known to over-reject severely, and the null-imposing Webb wild-cluster bootstrap we implement is known to under-reject (MacKinnon and Webb, 2018, 2020); consistent with this, it returns conservative p -values of 0.15 to 0.25. MacKinnon and Webb (2020) develop randomization inference for exactly this setting and find the t -statistic version the more reliable; Section 6.2 reports that studentized variant, and the upset-loss estimate clears the 5% level at three days under both permutation schemes. We therefore read

the analytic significance cautiously and rest the case on the broader pattern, the specificity to surprise, to the political domain, and to more-exposed respondents, rather than on any single test. The point estimate itself is stable, with no sign flips when any single province or event is dropped; Appendix Table A1 reports the wild-cluster bootstrap and the leave-one-out estimates.

Close losses show no comparable negative effect in any window. This contrast indicates that deviations from expectations, rather than losses per se, drive the result, consistent with reference-dependent utility (Kőszegi and Rabin, 2006). Upset wins are likewise insignificant, indicating that the spillover is asymmetric: negative surprises exert a stronger immediate effect than positive ones, in line with the well-documented negativity bias in affective responses (Baumeister et al., 2001).

Table 3: Effect of unexpected match outcomes on support for government non-interference.

	(1) Day 1	(2) Day 3	(3) Day 7
Upset loss	-0.294*** (0.090)	-0.224*** (0.054)	-0.149** (0.059)
Close loss	0.035 (0.091)	0.103 (0.136)	0.112* (0.059)
Upset win	-0.065 (0.065)	-0.060 (0.042)	-0.023 (0.040)
Mean of dep. var.	2.551	2.551	2.551
Province $\times$ year FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	35,114	35,114	35,114
R ²	0.057	0.058	0.058

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Upset loss, close loss, and upset win are counts of each outcome type in the pre-interview window, classified from football-data betting odds (Pinnacle closing); the omitted category comprises windows with only predicted outcomes or no match. All specifications include province $\times$ year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d - W, d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35,114$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Mechanism

The effect is domain-specific. Figure 3 plots the upset-loss coefficient, with 95% confidence intervals, for the main outcome and four placebo outcomes at each window. An upset loss reduces support for government non-interference with public criticism (reproducing the main estimate, -0.294 to -0.149 across windows), and no placebo outcome reproduces this profile: generalized social trust, life happiness, and non-interference in choices of work and residence are flat in every window, while non-interference in childbearing shows a weaker same-signed estimate that is marginal at seven days only (-0.136 , $p = 0.061$), without the sharp one-day response that marks the main effect. A diffuse worsening of mood would move trust and happiness; a generalized rise in support for government control would move both companion items at once; neither pattern appears. The concentration in the political domain instead fits the domain specificity of appraisal-driven emotions ([Lerner and Keltner, 2001](#)) and weighs against an account in which upset losses proxy for a broader provincial shock.

The effect is also concentrated among respondents more likely to have seen the result. An unexpected defeat of the provincial team is salient news that circulates widely online and in conversation, so the relevant margin is exposure rather than fandom. When we interact the shock with internet use, the closest CGSS proxy for connectedness to those information flows, the response is larger among high-internet respondents, consistent with an information-exposure channel (Appendix Table A3). This heterogeneity shares the few-treated-cluster limitation of the design and is read as directional rather than sharply estimated.

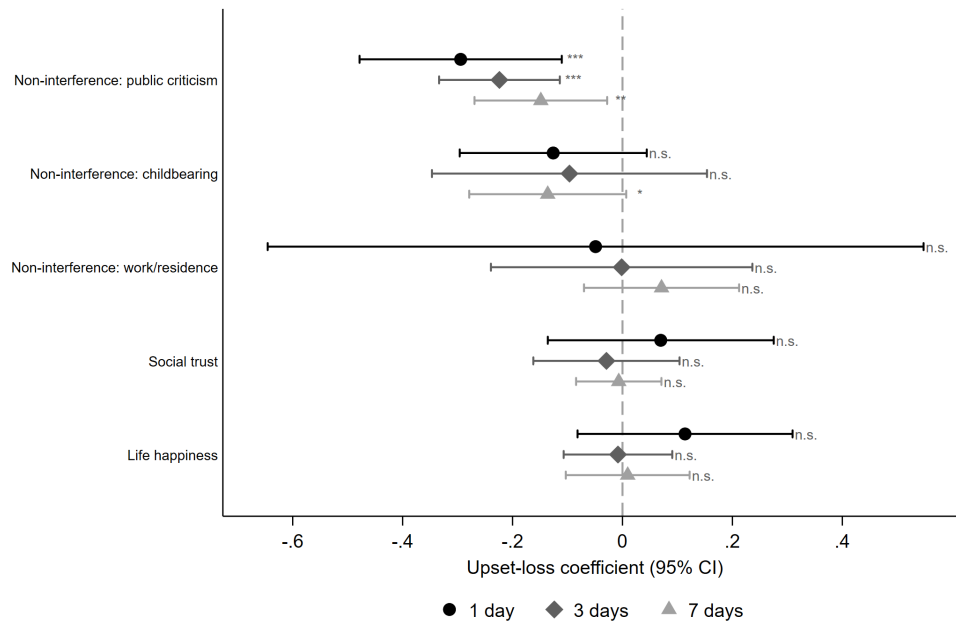


Figure 3: Domain specificity of the upset-loss effect.

Notes: The five outcomes are CGSS items, each on a 1–5 scale (higher values indicating more support for non-interference, or greater trust or happiness): three *Non-interference* items (public criticism, the main outcome; childbearing; work/residence) and two mood measures (social trust, life happiness). Each marker is the upset-loss coefficient, with its 95% confidence interval, from a separate regression per outcome and window under the Table 3 specification. Confidence intervals use province-clustered standard errors; CGSS weights.

5 Identification

This section probes the identifying assumption that exposure to an upset is unrelated to who is interviewed, through the timing of the effect and the balance of predetermined characteristics.

5.1 Event-Study Evidence

Figure 4 replaces the cumulative windows with binned indicators for upset losses occurring 1 to 3 and 4 to 7 days before the interview, alongside placebo bins for losses 1 to 3 and 4 to 7 days after, which should be zero under the identifying assumption. The 1-to-3-day pre-interview bin shows a significant negative effect (-0.197 , $p = 0.001$), while the 4-to-7-day bin remains negative but is not statistically significant. Both post-interview placebo bins are indistinguishable from zero, providing reassurance against pre-trends or reverse causality.

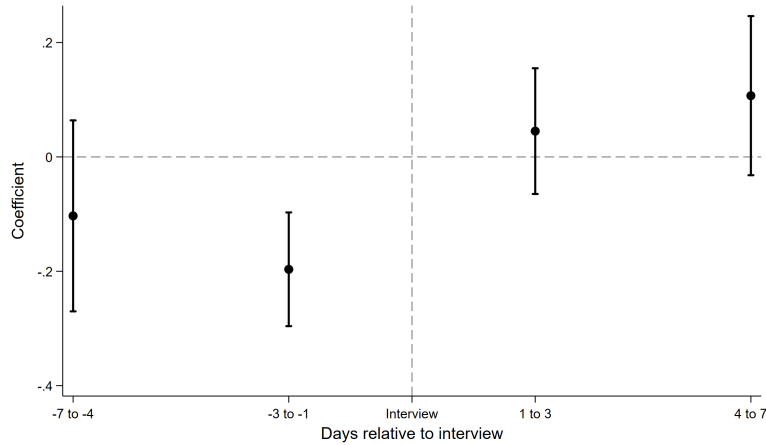


Figure 4: Event-study evidence: binned leads and lags of upset losses.

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each marker is the coefficient, with its 95% confidence interval, on a binned indicator for an upset loss relative to the interview date: negative event times are pre-interview windows (losses 1–3 and 4–7 days before the interview), positive event times post-interview placebo windows (losses 1–3 and 4–7 days after). All specifications include province $\times$ year fixed effects, individual, media, and Card predicted-category controls; CGSS weights ($N = 35,114$). Bars are 95% confidence intervals with province-clustered standard errors.

5.2 Balance and Composition

If exposure to an upset is unrelated to who is interviewed, respondents reached just before and just after an upset should be comparable. Figure 5 reports the before-versus-after comparison of [Muñoz et al. \(2020\)](#), plotting for each predetermined characteristic the standardized difference between respondents interviewed just after an upset (treated) and just before it (control, not yet exposed) within the three-day window, with its 95% confidence interval. Six of the seven characteristics are balanced (their intervals cross zero), and the seven are jointly insignificant ($F = 1.50$, $p = 0.374$ at three days, and $F = 0.51$, $p = 0.703$ at seven days). The exception is age: post-upset respondents are somewhat older. This gradient reflects older respondents being reached later in the fieldwork period rather than the upset itself; it is indistinguishable from zero at seven days and does not move the estimate once age is controlled.

Re-estimating the effect on this strict local sample reproduces the main result. Comparing support for non-interference just after versus just before an upset yields -0.220 at three days, essentially identical to the -0.224 in Table 3, so the estimate is not an artifact of the broader within-province-year control group on which the main specification relies.

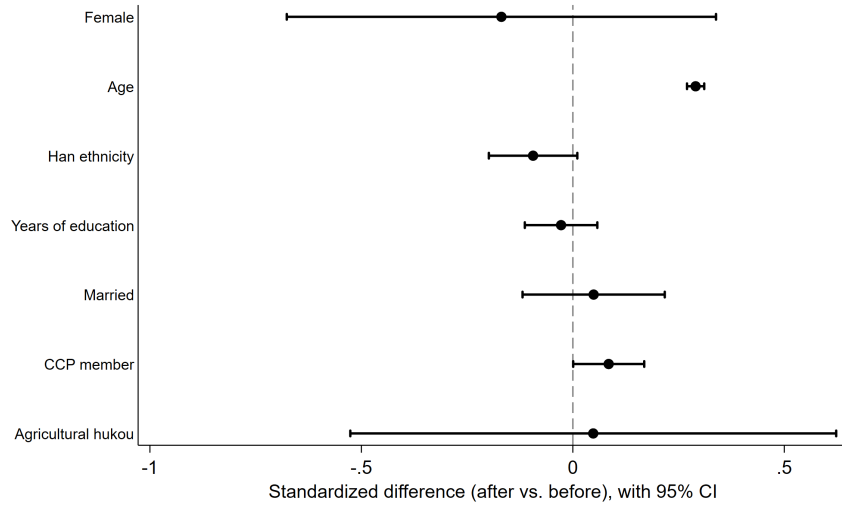


Figure 5: Balance of predetermined characteristics before and after an upset.

Notes: Canonical UESD design (Muñoz et al., 2020): within the three-day window around each upset, respondents interviewed just before it (not yet exposed) are the control and those just after are treated. Each marker is the standardized difference in a predetermined characteristic (after minus before, in standard-deviation units), with its 95% confidence interval, from a regression on the after indicator. Province $\times$ year fixed effects, province-clustered standard errors, CGSS weights.

6 Robustness and Inference

This section shows that the main estimate is stable across a wide range of alternative treatment definitions, control sets, samples, and fixed-effects structures, and probes inference with procedures designed for few treated clusters.

6.1 Robustness

Table 4 varies the definition of the treatment. Panel A redefines it as a single count or measure: an undifferentiated count of every loss produces no negative effect, and if anything a marginally positive one-day coefficient (0.127, $p < 0.10$), confirming that it is the surprise of an upset and not losing as such that moves attitudes, while a favorite-did-not-win count and a threshold-free continuous measure of favorite underperformance both reproduce the effect. Panel B raises the favorite threshold from 0.55 to 0.60 or 0.65, leaving the effect negative and, if anything, larger in magnitude, while the close-loss and upset-

win placebos remain mostly small and insignificant; the scattered exceptions are sign-inconsistent, led by a positive close-loss coefficient (0.255) in the one-day window at the 0.60 threshold, and no placebo reproduces the negative, decaying profile of the upset-loss effect. Figure 6 instead varies the controls and fixed effects: adding flexible interview-timing controls (day-of-week and month fixed effects and a holiday indicator), adding controls for the losing team's quality, form, ranking, home advantage, match importance, and finances, replacing the province-by-year fixed effects with separate province and year fixed effects plus province-level controls, and controlling for the exact home-match attendance respondents were exposed to in the days before the interview; each variation leaves the coefficient essentially unchanged. The effect also reflects the occurrence of an upset more than its margin of defeat (Appendix Table A2). Because the treatment is identified off few treated provinces, these checks speak to the stability of the point estimate rather than to its precision; we examine inference separately through the randomization inference reported below and the wild-cluster bootstrap and leave-out estimates in Appendix Table A1.

Table 4: Robustness to alternative treatment definitions.

<i>Panel A: Alternative treatment definitions (each block a separate single-treatment regression)</i>									
	Any loss (placebo)			Favorite did not win			Continuous		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Redefined treatment	0.127*	0.056	0.037	-0.455***	-0.202***	-0.100***	-0.445***	-0.239***	-0.134***
	(0.070)	(0.045)	(0.037)	(0.146)	(0.066)	(0.022)	(0.108)	(0.077)	(0.030)
<i>N</i>	35,114	35,114	35,114	35,114	35,114	35,114	35,114	35,114	35,114
<i>R</i> ²	0.057	0.057	0.057	0.057	0.058	0.058	0.057	0.057	0.057
<i>Panel B: Higher upset thresholds (each block a single joint regression)</i>									
	≥ 0.55 (baseline)			≥ 0.60			≥ 0.65		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.294***	-0.224***	-0.149**	-0.403***	-0.271***	-0.221***	-0.368***	-0.135	-0.182*
	(0.090)	(0.054)	(0.059)	(0.064)	(0.055)	(0.068)	(0.068)	(0.081)	(0.091)
Close loss	0.035	0.103	0.112*	0.255***	0.085	0.066	0.108*	-0.032	-0.013
	(0.091)	(0.136)	(0.059)	(0.065)	(0.071)	(0.060)	(0.060)	(0.049)	(0.044)
Upset win	-0.065	-0.060	-0.023	-0.042	-0.042	-0.035	-0.074	-0.076**	-0.028
	(0.065)	(0.042)	(0.040)	(0.055)	(0.041)	(0.045)	(0.058)	(0.035)	(0.047)
<i>N</i>	35,114	35,114	35,114	35,114	35,114	35,114	35,114	35,114	35,114
<i>R</i> ²	0.057	0.058	0.058	0.057	0.058	0.058	0.057	0.057	0.057

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Panel A replaces the upset-loss count with three single-treatment definitions: an any-loss count (every loss, odds ignored), a favorite-did-not-win count (a favorite with implied win probability at least 0.55 that lost or drew), and a continuous favorite-underperformance measure (the gap between the favorite's implied win probability and the realized result, on a probability scale). Panel B re-estimates the joint specification at favorite thresholds of 0.55 (baseline, as in Table 3), 0.60, and 0.65. All specifications include province $\times$ year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d - W, d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35,114$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

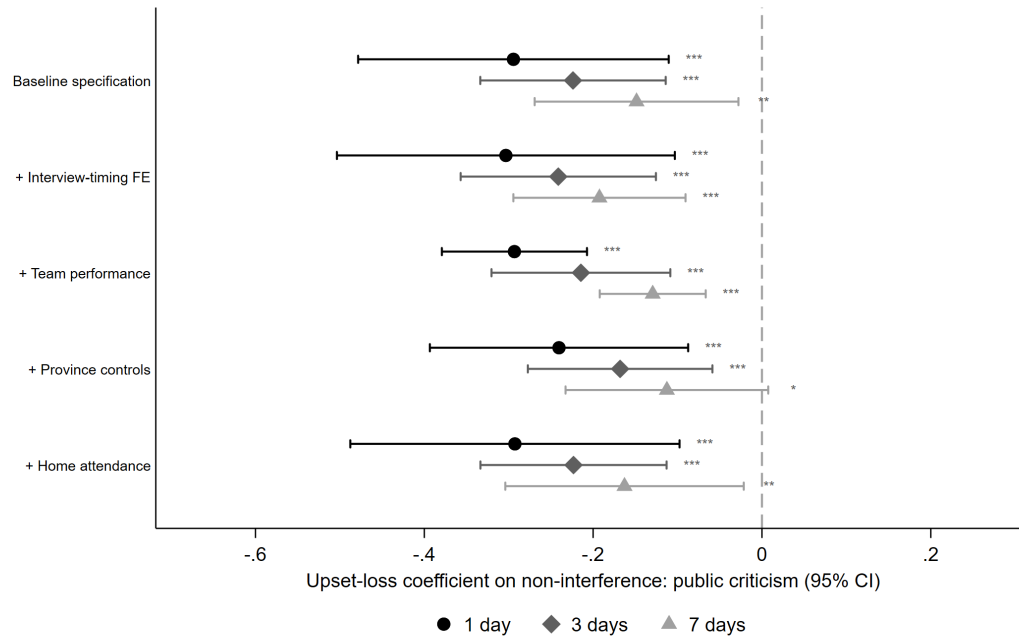


Figure 6: Robustness to alternative controls and fixed effects.

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each marker is the upset-loss coefficient, with its 95% confidence interval, at the one-, three-, and seven-day windows; the baseline reproduces Table 3 and each later specification adds to it: interview-timing controls (day-of-week and month fixed effects, a holiday indicator); team-performance controls (home status, form, standing, importance, favorite probability, transfer spending, attendance); province-level economic controls under separate province and year fixed effects; and exact home-match attendance (Transfermarkt). Close-loss and upset-win placebos are omitted for clarity; confidence intervals use province-clustered standard errors; CGSS weights ($N = 35,114$). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Randomization Inference

We assess statistical significance using randomization inference (Young, 2019; MacKinnon and Webb, 2020), permuting the full vector of match-outcome variables across province $\times$ date cells 5,000 times while holding covariates and fixed effects constant. The permutation is at the province $\times$ date level because that is the unit at which the treatment varies; the exercise therefore asks how often so few treated cells could produce estimates this large by chance, while the wild-cluster bootstrap of Appendix Table A1 addresses dependence at the broader province level, and the two procedures answer complementary questions. The p -value is the share of permutations producing a studentized coefficient (t -ratio) at least as extreme as observed, taken one-sided in the predicted direction for upset losses and two-sided for the close-loss and upset-win placebos.

Figure 7 plots the permutation distributions. The upset-loss coefficient falls in the left tail of its distribution in every window ($p = 0.053, 0.004$, and 0.022), while close-loss and upset-win coefficients sit near the center ($p = 0.850$ and 0.384 in the one-day window). A stricter variant that permutes the shock bundle only within province-by-year blocks, holding each province-year’s realized exposure fixed and randomizing only its timing across interview dates, yields $p = 0.139, 0.041$, and 0.069 , so the three-day estimate clears the 5% threshold under either randomization. Under both schemes the placebos remain central while upset losses sit in or near the left tail, a contrast that is difficult to attribute to chance alone.

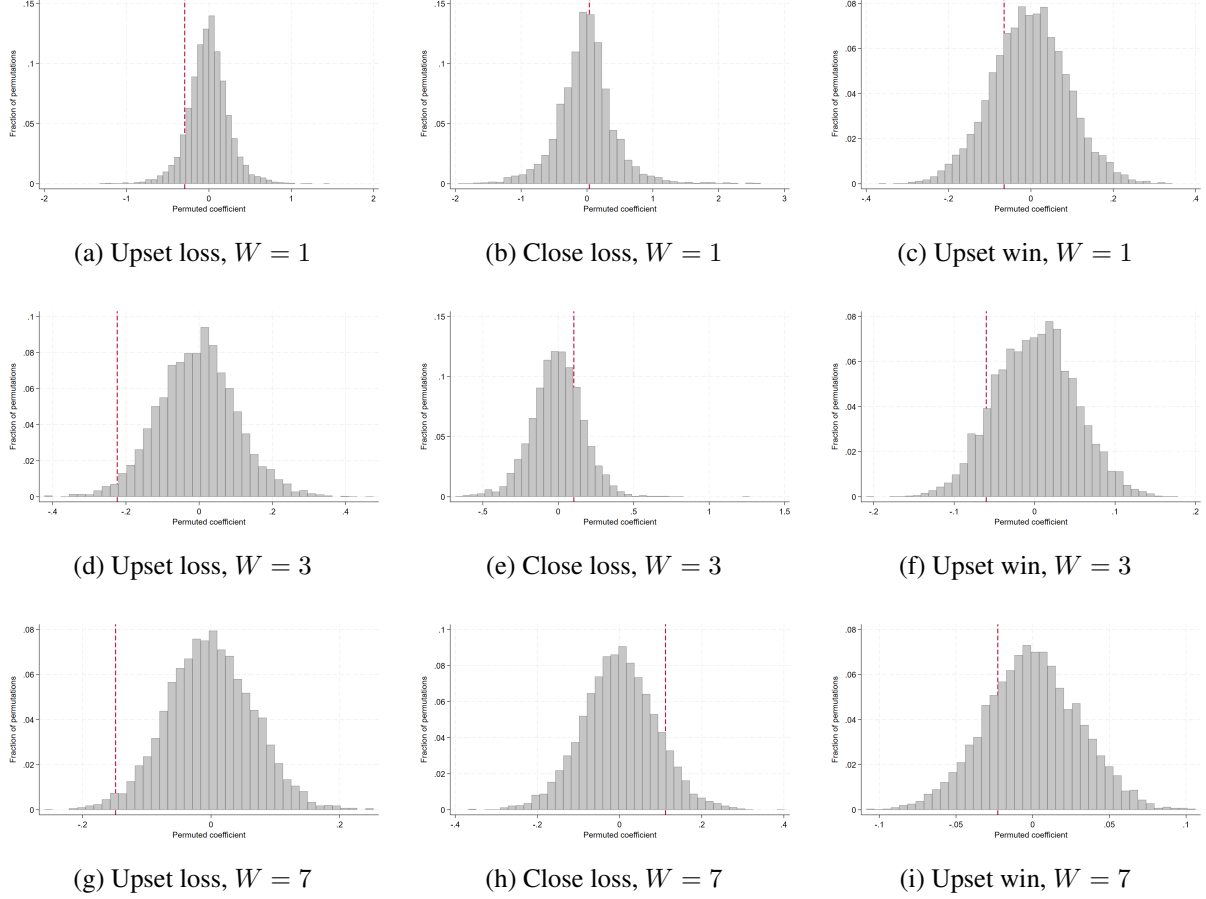


Figure 7: Randomization inference: permutation distributions of coefficients.

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each panel plots the coefficient distribution from 5,000 permutations of the match-outcome variables across province $\times$ date cells, with the red line at the actual estimate; rows are the 1-, 3-, and 7-day windows and columns the upset losses, close losses, and upset wins. All specifications include province $\times$ year fixed effects, individual, media, and Card predicted-category controls; CGSS weights ($N = 35,114$). The p -value is the share of permutations whose studentized coefficient is at least as extreme as observed, one-sided in the predicted direction for upset losses and two-sided for the placebos (Young, 2019).

7 Conclusion

Unexpected football losses temporarily increase public acceptance of government control over public criticism. An upset loss reduces support for non-interference with public criticism by 0.294 points (0.26 standard deviations) in the one-day window. The effect fades over the following week and is absent for close losses and upset wins.

Prior work on emotional spillovers from sports finds that where these shocks reach political outcomes, the response follows the valence of the shock: favorable surprises lift incumbent support and unfavorable ones erode trust in leaders, just as negative shocks produce violence, harsher sentencing, and stock market declines elsewhere. Our results show that this mapping reflects the outcome an emotion meets rather than a property of negative affect itself. Where the state is widely regarded as the guarantor of social order, transient negative arousal appears to increase demand for that order rather than erode trust in it. This response is consistent with the compensatory-control mechanism ([Kay et al., 2008](#)) and with non-sports evidence showing that incidental emotions shift judgments toward what the emotion makes people seek: protectionist and redistributive policies under induced negative emotions ([Algan et al., 2026](#)), and the status quo under rain-induced negative affect ([Meier et al., 2019](#)). The direction of the spillover is thus set by the outcome margin the emotion encounters and by the state's position as the provider of order, not by the valence of the shock alone.

These boundaries are best read as scope conditions on the mechanism rather than as a map of regime types. The pro-control turn should surface where the state is a salient and legitimate provider of order ([Kay et al., 2008, 2009](#)); where a shaken sense of control has some other ready outlet, the same defeat need not raise acceptance of state limits on public criticism. It also presumes an emotionally charged shock: the outcome has to be widely followed and bound up with local identification, so in settings where the sport means something different or draws a thinner following, both the affective jolt and its spillover

should fade. Finally, the sign we recover reflects a low-control, targetless appraisal; where incidental feeling is coupled more loosely to political judgment, or where a clear object of blame is at hand, the reaction could weaken or even reverse into a same-valence one (Healy et al., 2010).

Political attitudes are often treated as stable dispositions, yet our results show that they respond, at the margin, to emotional cues unrelated to government performance. Standard accounts of political support in China emphasize rational evaluation of government output (Fang et al., 2025) or durable dispositions transmitted across generations (Booth et al., 2022). Zhang et al. (2025) establish a causal link from life satisfaction to political trust in China, and Zhang and Churchill (2020) show that income inequality erodes subjective wellbeing partly through neighborhood trust, but in both cases the channel is a cognitive evaluation of life circumstances. Our results point to a more volatile component that moves with transient emotional states, and that may help explain short-run swings in public opinion otherwise difficult to attribute to policy changes.

Several limitations apply. First, our estimates are intent-to-treat effects at the province level, so the response among attentive fans may be substantially larger. Second, what we capture is a short-run movement in expressed attitudes that dissipates within a week, not a lasting revision of underlying political preferences. Third, with so few treated clusters, statistical significance varies across inference procedures. Randomization inference clears the 5% threshold at three days under both permutation schemes we report ($p = 0.004$ and 0.041), clears it at seven days under the baseline scheme only ($p = 0.022$ and 0.069), and is weakest in the one-day window, where upset losses are rarest; the wild-cluster bootstrap does not reach conventional thresholds, so significance should be read jointly with the specificity pattern (Section 6). Fourth, the CGSS is a repeated cross-section, so we cannot track within-individual changes or test whether repeated exposure produces cumulative effects. Finally, we cannot fully separate a purely affective reaction from one partly mediated by associations with football mismanagement, or from a re-

sponse to the wave of public venting a defeat can set off, in which case respondents react to the criticism they see around them rather than to their own arousal. The contribution, however, does not hinge on which proximate channel operates: a spillover from an apolitical sporting outcome to political attitudes is the phenomenon we document; the effect is concentrated in tolerance of public criticism rather than reflecting a generalized demand for state control (Figure 3); and the treatment is the unexpected defeat of a provincial club rather than the national team, the usual focus of football-corruption discourse. Relatedly, although the surprise of the result and the absence of a blameworthy target point to a low-control, uncertainty appraisal rather than one organized around blame ([Smith and Ellsworth, 1985](#); [Lerner and Keltner, 2001](#)), survey data cannot isolate which specific negative emotion within that profile (anxiety, frustration, or disappointment) carries the effect; direct measurement of affect could resolve this. Future work with panel data or real-time survey designs could address these limitations.

References

- Yann Algan, Eva Davoine, Thomas Renault, and Stefanie Stantcheva. Emotions and policy views. Harvard University Working Paper, 2026. URL <https://socialeconomicslab.org/research/working-papers/emotions-and-policy/>.
- Leander Andres, Marc Fabel, and Helmut Rainer. How much violence does football hooliganism cause? *Journal of Public Economics*, 225:104970, 2023. <https://doi.org/10.1016/j.jpubeco.2023.104970>.
- M. Niaz Asadullah, Saizi Xiao, and Emile Yeoh. Subjective well-being in China, 2005–2010: The role of relative income, gender, and location. *China Economic Review*, 48:83–101, 2018. <https://doi.org/10.1016/j.chieco.2015.12.010>.
- Roy F. Baumeister, Ellen Bratslavsky, Catrin Finkenauer, and Kathleen D. Vohs. Bad is stronger than good. *Review of General Psychology*, 5(4):323–370, 2001. <https://doi.org/10.1037/1089-2680.5.4.323>.
- Alison Booth, Xin Meng, Elliott Fan, and Dandan Zhang. The direct and intergenerational behavioural consequences of a socio-political upheaval. *Journal of Economic Behavior & Organization*, 200:931–958, 2022. <https://doi.org/10.1016/j.jebo.2022.06.033>.
- Qingyin Cai and Qingxiao Li. Emotional shocks and consumer spending. *Journal of Economic Behavior & Organization*, 241:107357, 2026. <https://doi.org/10.1016/j.jebo.2025.107357>.
- David Card and Gordon B. Dahl. Family violence and football: The effect of unexpected emotional cues on violent behavior. *The Quarterly Journal of Economics*, 126(1):103–143, 2011. <https://doi.org/10.1093/qje/qjr001>.
- Alexander Cardazzi, Bryan C. McCannon, Brad R. Humphreys, and Zachary Rodriguez. Emotional cues and violent behavior: Unexpected basketball losses increase incidents of family violence. *The Journal of Law, Economics, and Organization*, 40(2):267–288, 2024. <https://doi.org/10.1093/jleo/ewac014>.
- Dan Chen and Andrew W. MacDonald. Bread and circuses: Sports and public opinion in China. *Journal of Experimental Political Science*, 7(1):41–55, 2020. <https://doi.org/10.1017/XPS.2019.15>.
- Jidong Chen, Jennifer Pan, and Yiqing Xu. Sources of authoritarian responsiveness: A field experiment in China. *American Journal of Political Science*, 60(2):383–400, 2016. <https://doi.org/10.1111/ajps.12207>.
- Michael Church. Chinese transfer record broken five times in 2016 as Oscar arrives. ESPN, December 24, 2016. https://www.espn.com/soccer/story/_/id/37463836/chinese-transfer-record-broken-five-s-2016-oscar-arrives.
- Daniel Cockayne, Simon Chadwick, and Jonathan Sullivan. Chinese football – from a state-led past to a digital future. *Journal of Global Sport Management*, 7(3):345–354, 2022. <https://doi.org/10.1080/24704067.2021.1883212>.
- Decio Coviello, Erika Deserranno, Nicola Persico, and Paola Sapienza. Effect of mood and worker incentives on workplace productivity. *The Journal of Law, Economics, and Organization*, 40(2):362–393, 2024. <https://doi.org/10.1093/jleo/ewac017>.

- Edward Cunningham, Tony Saich, and Jesse Turiel. Understanding CCP resilience: Surveying Chinese public opinion through time. Ash Center policy brief, Ash Center for Democratic Governance and Innovation, Harvard Kennedy School, Cambridge, MA, 2020.
- Emilio Depetris-Chauvin, Ruben Durante, and Filipe Campante. Building nations through shared experiences: Evidence from African football. *American Economic Review*, 110(5):1572–1602, 2020. <https://doi.org/10.1257/aer.20180805>.
- Alex Edmans, Diego García, and Øyvind Norli. Sports sentiment and stock returns. *The Journal of Finance*, 62(4):1967–1998, 2007. <https://doi.org/10.1111/j.1540-6261.2007.01262.x>.
- Ozkan Eren and Naci Mocan. Emotional judges and unlucky juveniles. *American Economic Journal: Applied Economics*, 10(3):171–205, 2018. <https://doi.org/10.1257/app.20160390>.
- Ming Fang, Weizheng Lai, and Congling Xia. Anti-corruption and political trust: Evidence from China. *Journal of Economic Behavior & Organization*, 234:107015, 2025. <https://doi.org/10.1016/j.jebo.2025.107015>.
- Stanley Feldman and Karen Stenner. Perceived threat and authoritarianism. *Political Psychology*, 18(4): 741–770, 1997. <https://doi.org/10.1111/0162-895X.00077>.
- Martin Fochmann, Frank Hechtner, Erich Kirchler, and Peter N.C. Mohr. When happy people make society unhappy: Emotions affect tax compliance behavior. *Journal of Economic Behavior & Organization*, 229:106854, 2025. <https://doi.org/10.1016/j.jebo.2024.106854>.
- Anthony Fowler and B. Pablo Montagnes. College football, elections, and false-positive results in observational research. *Proceedings of the National Academy of Sciences*, 112(45):13800–13804, 2015. <https://doi.org/10.1073/pnas.1502615112>.
- Qi Ge. Sports sentiment and tipping behavior. *Journal of Economic Behavior & Organization*, 145: 95–113, 2018. <https://doi.org/10.1016/j.jebo.2017.10.016>.
- General Office of the State Council. Notice on issuing the Overall Plan for Chinese Football Reform and Development. Guobanfa [2015] No. 11, March 16, 2015. https://www.gov.cn/zhengce/content/2015-03/16/content_9537.htm.
- Mengmeng Guo, Dian Ling, and Qiang Wen. Female Olympic champions and gender preferences: Evidence from China. *Journal of Economic Behavior & Organization*, 240:107299, 2025. <https://doi.org/10.1016/j.jebo.2025.107299>.
- Andrew J Healy, Neil Malhotra, and Cecilia Hyunjung Mo. Irrelevant events affect voters’ evaluations of government performance. *Proceedings of the National Academy of Sciences*, 107(29):12804–12809, 2010. <https://doi.org/10.1073/pnas.1007420107>.
- William Hennelly. China’s soccer attendance catching up to European giants’. China Daily, January 21, 2016. http://www.chinadaily.com.cn/world/2016-01/21/content_23179108.htm.
- Ria Ivandić, Tom Kirchmaier, Yasaman Saeidi, and Neus Torres Blas. Football, alcohol, and domestic abuse. *Journal of Public Economics*, 230:105031, 2024. <https://doi.org/10.1016/j.jpubeco.2023.105031>.

- Hemant Kakkar and Niro Sivanathan. When the appeal of a dominant leader is greater than a prestige leader. *Proceedings of the National Academy of Sciences*, 114(26):6734–6739, 2017. <https://doi.org/10.1073/pnas.1617711114>.
- Aaron C. Kay, Danielle Gaucher, Jamie L. Napier, Mitchell J. Callan, and Kristin Laurin. God and the government: Testing a compensatory control mechanism for the support of external systems. *Journal of Personality and Social Psychology*, 95(1):18–35, 2008. <https://doi.org/10.1037/0022-3514.95.1.18>.
- Aaron C. Kay, Jennifer A. Whitson, Danielle Gaucher, and Adam D. Galinsky. Compensatory control: Achieving order through the mind, our institutions, and the heavens. *Current Directions in Psychological Science*, 18(5):264–268, 2009. <https://doi.org/10.1111/j.1467-8721.2009.01649.x>.
- Judd B. Kessler, Andrew McClellan, James Nesbit, and Andrew Schotter. Short-term fluctuations in incidental happiness and economic decision-making: Experimental evidence from a sports bar. *Experimental Economics*, 25(1):141–169, 2022. <https://doi.org/10.1007/s10683-021-09708-9>.
- Kyosuke Kikuta and Mamoru Uesugi. Do politically irrelevant events cause conflict? The cross-continental effects of European professional football on protests in Africa. *International Organization*, 77(1):179–216, 2023. <https://doi.org/10.1017/S0020818322000261>.
- Gary King, Jennifer Pan, and Margaret E. Roberts. How censorship in China allows government criticism but silences collective expression. *American Political Science Review*, 107(2):326–343, 2013. <https://doi.org/10.1017/S0003055413000014>.
- Botond Köszegi and Matthew Rabin. A model of reference-dependent preferences. *The Quarterly Journal of Economics*, 121(4):1133–1165, 2006. <https://doi.org/10.1093/qje/121.4.1133>.
- Mark J. Landau, Aaron C. Kay, and Jennifer A. Whitson. Compensatory control and the appeal of a structured world. *Psychological Bulletin*, 141(3):694–722, 2015. <https://doi.org/10.1037/a0038703>.
- Jennifer S Lerner and Dacher Keltner. Fear, anger, and risk. *Journal of Personality and Social Psychology*, 81(1):146–159, 2001. <https://doi.org/10.1037/0022-3514.81.1.146>.
- Jennifer S Lerner, Ye Li, Piercarlo Valdesolo, and Karim S Kassam. Emotion and decision making. *Annual Review of Psychology*, 66(1):799–823, 2015. <https://doi.org/10.1146/annurev-psych-010213-115043>.
- James G. MacKinnon and Matthew D. Webb. The wild bootstrap for few (treated) clusters. *The Econometrics Journal*, 21(2):114–135, 2018. <https://doi.org/10.1111/ectj.12107>.
- James G. MacKinnon and Matthew D. Webb. Randomization inference for difference-in-differences with few treated clusters. *Journal of Econometrics*, 218(2):435–450, 2020. <https://doi.org/10.1016/j.jeconom.2020.04.024>.
- Armando N. Meier, Lukas Schmid, and Alois Stutzer. Rain, emotions and voting for the status quo. *European Economic Review*, 119:434–451, 2019. <https://doi.org/10.1016/j.euroecorev.2019.07.014>.
- Jordi Muñoz, Albert Falcó-Gimeno, and Enrique Hernández. Unexpected event during survey design: Promise and pitfalls for causal inference. *Political Analysis*, 28(2):186–206, 2020. <https://doi.org/10.1017/pan.2019.27>.

- Andrew J. Nathan. Authoritarian resilience. *Journal of Democracy*, 14(1):6–17, 2003. <https://doi.org/10.1353/jod.2003.0019>.
- Francesco Passarelli and Guido Tabellini. Emotions and political unrest. *Journal of Political Economy*, 125(3):903–946, 2017. <https://doi.org/10.1086/691700>.
- Daniel I. Rees and Kevin T. Schnepel. College football games and crime. *Journal of Sports Economics*, 10(1):68–87, 2009. <https://doi.org/10.1177/1527002508327389>.
- Darrel Robinson and Marcus Tannenberg. Self-censorship of regime support in authoritarian states: Evidence from list experiments in China. *Research & Politics*, 6(3):1–9, 2019. <https://doi.org/10.1177/2053168019856449>.
- Norbert Schwarz and Gerald L Clore. Mood, misattribution, and judgments of well-being: Informative and directive functions of affective states. *Journal of Personality and Social Psychology*, 45(3):513–523, 1983. <https://doi.org/10.1037/0022-3514.45.3.513>.
- Craig A. Smith and Phoebe C. Ellsworth. Patterns of cognitive appraisal in emotion. *Journal of Personality and Social Psychology*, 48(4):813–838, 1985. <https://doi.org/10.1037/0022-3514.48.4.813>.
- Liam Twomey. Chelsea announce £60 million Oscar transfer to Shanghai SIPG. ESPN, December 23, 2016. https://www.espn.com/soccer/story/_/id/37507224.
- Xinhua. Chinese Super League sells TV rights in record 8bn yuan. China Daily, October 28, 2015. https://www.chinadaily.com.cn/sports/2015-10/28/content_22303384.htm.
- Yu You, Yifan Huang, and Yuyi Zhuang. Natural disaster and political trust: A natural experiment study of the impact of the Wenchuan earthquake. *Chinese Journal of Sociology*, 6(1):140–165, 2020. <https://doi.org/10.1177/2057150X19891880>.
- Alwyn Young. Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *The Quarterly Journal of Economics*, 134(2):557–598, 2019. <https://doi.org/10.1093/qje/qjy029>.
- Quanda Zhang and Sefa Awaworyi Churchill. Income inequality and subjective wellbeing: Panel data evidence from China. *China Economic Review*, 60:101392, 2020. <https://doi.org/10.1016/j.chieco.2019.101392>.
- Youxing Zhang, Peter Howley, and Clemens Hetschko. Happy citizens trust their rulers. *Journal of Population Economics*, 38(3):65, 2025. <https://doi.org/10.1007/s00148-025-01120-4>.

Appendix

A. Inference with Few Treated Clusters

Table A1 collects the inference for the upset-loss coefficient given the few treated provinces: the analytic province-clustered p -value, the Webb wild-cluster bootstrap, and leave-one-province-out and leave-one-event-out estimates; randomization inference is reported in Section 6.2. The point estimate is robust to dropping any single province or event, while the wild-cluster bootstrap p -values are larger, as the null-imposing wild-cluster bootstrap is expected to under-reject with so few treated clusters (MacKinnon and Webb, 2018).

Table A1: Inference for the upset-loss effect with few treated clusters.

	Day 1	Day 3	Day 7
Upset-loss coefficient	−0.294	−0.224	−0.149
clustered standard error	(0.090)	(0.054)	(0.059)
analytic p -value	0.003	0.000	0.018
wild-cluster bootstrap p	0.202	0.146	0.249
Leave-one-province-out range	[−0.42, −0.19]	[−0.25, −0.17]	[−0.18, −0.08]
sign flips; worst p	0; 0.003	0; 0.019	0; 0.219
Leave-one-event-out range	[−0.38, −0.19]	[−0.33, −0.15]	[−0.17, −0.11]
sign flips; worst p	0; 0.165	0; 0.052	0; 0.114

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference); all entries are for the upset-loss coefficient under the Table 3 specification. Each column reports the coefficient with its clustered standard error, the analytic and Webb wild-cluster-bootstrap p -values, and leave-one-province-out and leave-one-event-out coefficient ranges (each treated province, respectively each treated province $\times$ date event, dropped in turn); sign flips counts the re-estimations with a positive coefficient. Football-data odds (Pinnacle closing), CGSS 2015–2023; province-clustered standard errors; CGSS sampling weights.

B. Additional Robustness

Appendix Table A2 adds the cumulative goal deficit of the upset losses, separating the occurrence of an upset from its severity.

Table A2: Upset-loss severity: occurrence versus margin of defeat.

	Day 1	Day 3	Day 7
Upset loss (count)	-0.485*** (0.095)	-0.607*** (0.220)	-0.274** (0.116)
Upset-loss goal deficit	0.090 (0.102)	0.185** (0.069)	0.070 (0.042)
Close loss	0.035 (0.091)	0.106 (0.136)	0.110* (0.061)
Upset win	-0.070 (0.065)	-0.074* (0.040)	-0.031 (0.039)
Mean of dep. var.	2.551	2.551	2.551
Province $\times$ year FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	35,114	35,114	35,114
R^2	0.057	0.058	0.058

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). The specification adds the cumulative goal deficit of the upset losses (the margin of defeat) alongside the upset-loss count; close loss and upset win enter as counts. All specifications include province $\times$ year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d - W, d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35,114$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C. Heterogeneity

Table A3 reports formal interaction tests of heterogeneity by internet use, agricultural hukou, and gender. No upset-loss interaction is significant in all three windows simultaneously, so the heterogeneity is read as suggestive and underpowered rather than as identifying which respondents drive the effect. The two margins that do reach significance also point in different directions: high internet use amplifies the response in the shortest windows, while agricultural hukou amplifies it at the three- and seven-day horizons even though rural respondents are on average less connected. This tension is further reason not to read the heterogeneity as isolating a single exposure channel.

Table A3: Heterogeneity in the upset-loss effect.

	High internet use			Agricultural hukou			Female		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.134 (0.159)	-0.134** (0.059)	-0.107*** (0.030)	-0.244 (0.213)	-0.151* (0.077)	-0.088 (0.071)	-0.311*** (0.113)	-0.361*** (0.116)	-0.098 (0.103)
Interaction	-0.328* (0.182)	-0.179** (0.083)	-0.078 (0.093)	-0.085 (0.271)	-0.214*** (0.059)	-0.241* (0.124)	0.035 (0.063)	0.309 (0.190)	-0.111 (0.120)
Close loss	0.061 (0.090)	0.130 (0.174)	0.151** (0.059)	0.990** (0.447)	0.451** (0.170)	0.203** (0.092)	0.328*** (0.091)	0.207 (0.138)	0.104 (0.081)
Interaction	-0.382*** (0.034)	-0.110 (0.196)	-0.115 (0.199)	-1.113** (0.446)	-0.606*** (0.074)	-0.166 (0.110)	-0.598*** (0.045)	-0.225*** (0.038)	0.016 (0.084)
Upset win	-0.002 (0.105)	-0.028 (0.044)	-0.010 (0.049)	-0.134 (0.126)	-0.099 (0.071)	-0.025 (0.044)	-0.117 (0.090)	-0.105* (0.052)	-0.025 (0.057)
Interaction	-0.153 (0.120)	-0.086 (0.066)	-0.036 (0.073)	0.110 (0.162)	0.053 (0.072)	-0.006 (0.036)	0.102 (0.108)	0.082* (0.048)	0.005 (0.049)
Province $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Media controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	35,114	35,114	35,114	35,114	35,114	35,114	35,114	35,114	35,114
R^2	0.057	0.058	0.058	0.057	0.058	0.058	0.057	0.058	0.058

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each three-column block is one full-sample regression interacting every shock with the block's subgroup indicator (high internet use, agricultural hukou, or female); the indented Interaction row is the coefficient on the interaction and the shock row is the effect in the omitted category, so the subgroup's effect is the sum of the two. All specifications include province $\times$ year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d - W, d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35,114$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.